

BPRO Final Project: Golden Record V2

ACTUAL PROJECT CLICK HERE: [LINK TO PROJECT WEBSITE](#)

Written Portion

Ha's Section:

The Voyager Golden records are two phonographs containing sounds and images chosen to portray life and culture on Earth, sent into space as a message in a glass bottle for intelligent extraterrestrial life to find and interpret. The records were curated by a team led by Carl Sagan and represent a time capsule of humanity's scientific understanding and cultural richness at a particular moment in history (1977). It has been almost fifty years since then, and much has changed in our knowledge of fundamental physics and in the breadth and diversity of human culture we are now better equipped to represent. This document details the contents of our interactive website which serves as a proxy for a new Voyager Golden Record, updated for the 21st century, along with the reasoning behind each selection. For the sake of clarity in this assignment, we are simply opting to use English as the language of communication; considerations of how English may be rendered legible to extraterrestrial intelligence is beyond the scope of this project.

I. Numbers and Conversions:

Any meaningful communication with an extraterrestrial intelligence must begin with a shared logical foundation. Mathematics is our best candidate for a universal language—as Galileo famously said: “Mathematics is the alphabet with which God has written the universe”. We therefore begin our record with a primer on number systems.

Like in the movie *Contact*, we provide a list of prime numbers up until 10^3 . These ultimately quite rudimentary, but universally important, numbers provide the basis on which extraterrestrial intelligence can calibrate their own numerical system(s) to.

We then provide explicit conversion tables between three number bases: binary (base-2), decimal (base-10), and duodecimal (base-12). Binary is included as the foundational language of modern computing and digital information processing; it is the system in which all data on this website is ultimately encoded, and any civilisation capable of receiving it may also have based their technologies on binary due to its simplicity and easy assignment to on and off states of electrical current. Decimal is included as the most widely used number system among humans, arising in part from our ten fingers, and it underlies almost all further math and science communicated in this document. Duodecimal is included as a mathematically elegant base, divisible by 2, 3, 4, and 6, which some argue represents a more natural counting system for the purposes of division and fractions. It provides an alternative to our default preference for base-10, which is more anatomical rather than mathematically pure. We anticipate that the primary counting system of an alien species is also, to a degree, anatomically influenced.

Alongside these conversion tables, we include a list of fundamental physical constants expressed in base-10 SI units. These constants, including the speed of light in a vacuum, Planck's constant,

the gravitational constant, the elementary charge, etc are empirically measurable features of the universe itself. They form the basis for many fundamental physics equations and serve as a bridge to the following section on physics.

II. Universal Physics:

While less stable than our mathematics and oftentimes incomplete, our physical theories should also be universal (even incomplete theories should partially overlap with incomplete theories that extraterrestrial intelligence may have, if they both claim universality). These universals, again, can be considered a Rosetta Stone for extraterrestrial intelligence to begin decoding this information that we are sending out to them. The following webpages include important physics theories and can serve as a handshake for extraterrestrial civilizations reading the record to understanding our state of scientific advancement, as well as a basis for further communicating chemistry, geology, and biology:

1. Λ CDM

Our most successful cosmological model so far is of a universe comprising a nearly flat spacetime whose expansion rate is determined by cold gravitating matter and a constant vacuum energy density (Λ), with the latter currently dominating and causing accelerated expansion. It is known as the Λ CDM cosmological model. Advanced alien civilisations would, presumably, have developed their own cosmological models that may in part overlap with Λ CDM. Including theory and key equations in Λ CDM is thus crucial to communicating our level of understanding of the universe. (*Lecture 15*)

2. Timeline of the Universe

As predicted by the Lambda CDM cosmological model, the Universe would have started with the Big Bang and inflated through a number of different eras to the dark-energy-dominated, star and galaxy speckled Universe that we have today. If nothing else, this history is something that we surely share with any alien species out there. (*Lecture 15*)

3. The Standard Model

The Standard Model is the quantum field theory that describes the fundamental particle and fundamental forces, bar gravity, in our universe. It classifies all known elementary particles into two broad families: fermions, which make up baryonic matter, and bosons, which mediate the fundamental forces. The Standard Model is, empirically, among the most accurate scientific theories ever formulated. These particles and forces it describes accounts for all matter and interactions in the observable universe; given our own assuredness in the Standard Model, any equivalent and successful alien quantum field theory should overlap with the Standard Model by a wide margin.

4. The Periodic Table

The periodic table organizes all known chemical elements (baryonic matter) by increasing atomic number and reveals the recurring patterns in their chemical properties that arise from the quantum mechanical structure of electron orbitals. It currently contains 118 confirmed elements and its structure is now understood as a direct consequence of the same quantum rules described in the previous section.

From our understanding of their production in stars, the same elements should exist throughout the observable universe and as such the periodic table serves as a chemical alphabet between us and extraterrestrial intelligence, allowing them to understand further concepts that we will attempt to communicate in the subsequent sections beyond physics, like chemistry, geology, and biology.

5. **Galaxies, Stars, and Exoplanets**

We include a classification of galaxy morphologies (elliptical, spiral, lenticular, and irregular) and describe the large-scale structure of the universe.

For stars, we include a video tutorial explaining the Hertzsprung-Russell diagram, which organises stars by luminosity and surface temperature and reveals the main sequence of hydrogen-burning stars alongside red giants, white dwarfs, and other evolutionary endpoints (neutron stars and black holes). We describe the stellar lifecycle, and the production of elements in stars via an interactive 2048-esque game. Likewise, for planets, we describe the variety of planetary types, from hot Jupiters to rocky terrestrial planets and include simulations of protoplanetary disk formation in nebulae and the formation of the Solar System, which leads us into our next section. (*Lecture 6, Lecture 8*)

III. **The Earth as a Living Planet:**

The Earth as we have learnt is not so special a planet in its inorganic content and structure, but is unique in its role as a host of life. These following webpages provide information on our planet, from its formation to the wealth of biodiversity and the Anthropocene in which we find ourselves today.

1. **Solar System and planet formation**

Following lecture 8, we show the main bodies within our solar system, their classification, as well as a 3D visualization of the Solar System from NASA, such that we may easily be identified if extraterrestrial life stumble into our local neighborhood (*Lecture 8*)

<https://eyes.nasa.gov/apps/solar-system/#/home>

2. **Evolution and Tree of life:**

<https://www.onezoom.org/life/@biota=93302?vis=polytomy&otthome=%40ozid%3D1#x762,y1311,w2.2515>

We include the linked website, which comprehensively charts the evolutionary relation between life (as we are currently able to classify it) on Earth, as well as a brief timeline of the evolution of life on our planet (*Lecture 2*).

3. **List of Biosignatures from Earth and spectra**

Are extraterrestrial species searching for us as much as we have been actively searching for them, via biosignatures? We provide a list of detectable biosignatures in our atmosphere, as well as a sample spectrum that they can reference (*Lecture 9*)

4. **Images and sounds on Earth**

Here, we choose to maintain the original Voyager Golden Record recordings, encapsulating a broad range of sounds of nature on Earth

https://soundcloud.com/nasa/sets/golden-record-sounds-of?utm_source=clipboard&utm_campaign=wtshare&utm_medium=widget&utm_content=https%253A%252F%252Fsoundcloud.com%252Fnasa%252Fsets%252Fgolden-record-sounds-of

IV. Humans & the Anthropocene:

Course content has focused heavily on the physical and biological origins and functions of human life and how these scientific bases may generalize to extraterrestrial life, but not so much on society and what our understanding of human civilization can generalize to extraterrestrial civilizations that, in the case they are able to intercept and interpret this Golden Record, must be quite advanced. In the following pages, we aim to represent both the human body and the human mind at the individual and collective level.

1. Human anatomy:

The human body is the basis of all of our functions, and communicating this anatomy is crucial for extraterrestrial intelligence to begin to understand the very basic foundation of what it means to be human. We include media from the NIH's Visible Human project, which shows in detail cross sections of the male and female human body.

<https://www.youtube.com/watch?v=7GPB1sjEhIQ>

https://www.nlm.nih.gov/research/visible/mpeg/umd_video.mpg

2. Human history:

This is an incredibly complex topic, and we have included a short (and humorous) overview of human history here, covering major civilizations, religions, power systems, conflicts, and achievements.

<https://www.youtube.com/watch?v=xuCn8ux2gbs>

3. Greetings & languages

The original golden record included welcoming greetings in 100+ languages. Since then, great anthropological and linguistic measures have been made. We expand our communication capability by including *No Language Left Behind*, an Meta AI multilingual translation model trained on data using data mining techniques tailored for low-resource languages and supporting over 200 languages.

<https://ai.meta.com/research/no-language-left-behind/>

<https://huggingface.co/facebook/nllb-200-3.3B>

4. Arts

The ability to create art is something unique to humans (*Lecture 18*). We include here some examples of our artistic creations in 4 different media:

- Visual art:

We include <https://artsandculture.google.com/>, which provides a wealth of visual arts—our oldest art form.

- Film:

The film we chose is Wall-E. For obvious reasons, we can't actually have the full film on the website, but we selected this film for its empathetic reflection of the failures of human society (climate change, pollution), but also our capacity for advanced technological advancements and our ability to connect with non-human, non-organic beings.

- Music

We include the original samples of songs that the Voyager Golden Record compiled, spanning a variety of genres. To add further diversity and fill in the gap that the original record left with regard to contemporary music, we

will include a Spotify playlist containing songs from every country in the world and territories.

<https://open.spotify.com/playlist/4Gpwujj4eIY1Ypop8SL8cl?trackId=6q3wafztoGDT3nQYWVfyuO>

- Video games:

Humans, and many other forms of life on Earth, love to play, and how better can we share our games with extraterrestrial beings than through digital format. We include here [Little Alchemy](#) as one example of a video game, and link to itch.io, a repository of independently developed video games that can be explored, demonstrating the range of human creativity.

5. **Technology and Science:**

We provide an overview of the major scientific advancements that we have made as a species, beginning with units of measurement in the Bronze Age to the sequencing of a full human gene in 2022.

https://en.wikipedia.org/wiki/Timeline_of_scientific_discoveries

We also include a list of technosignatures one could possibly use to detect Earth:

<https://www.seti.org/news/detecting-earth-how-far-away-can-we-detect-earth-s-technosignatures/>

(Lecture 10)

6. **Knowledge bases:**

At the end of the day, there is so much more knowledge that cannot be encapsulated within this project. We provide two interactive databases that can be explored:

<https://www.nature.com/immersive/d41586-019-03165-4/index.html>

This contains 150 years of rigorously peer-reviewed and prestigious research, visualised by impact.

<https://wikimap.wiki/?base=map&lat=-3698329.1765&lon=3091724.9201&showAll=true&wiki=enwiki&zoom=3>

It is equally important to highlight the process of community-based knowledge compilation and free access to knowledge, and there is no database more accessible and extensive than Wikipedia. This website allows an exploration of Wikipedia that links different topics by their relevance to each other.

Rhea's Section:

The Drake Equation, proposed by Frank Drake at the 1961 Green Bank conference, is what inspired this portion of our project. As discussed in Lecture 1 (Serenio) and referenced in Frank et al. (2023), the equation was designed not to produce a definitive answer but to identify and quantify the key unknowns governing the prevalence of intelligent and communicating civilizations. Each variable spans a different scientific domain, which made me think of its compatibility as a multi-pad synthesizer: each pad represents a different category of scientific uncertainty.

Here are my justifications and thought processes for the sonic mapping chosen for each variable in the Drake Equation. The least justified setting is the base frequency: 432 Hz, which is

colloquially known as the “frequency of the universe” in some cultures and religions. Because of the lack of scientific grounding for this frequency, I made sure that the base frequency was editable. This is the only arbitrary variable that is not directly attached to the Drake Equation.

1) $R_{\star}$ — Star Formation Rate (Astrophysics)

- a) $R_{\star}$ is the rate at which new stars form in the Milky Way, in solar masses per year. I learned that it is the first term because all other terms depend on the existence of stellar systems. Current observational estimates from infrared and radio surveys converge on 1–3 solar masses per year, so the default in my system is set to 1.5. The star formation rate has declined since its cosmic peak approximately 10 billion years ago, which is the scientific justification for the modulation feature: sweeping $R_{\star}$ over time models how the baseline opportunity for life has changed across cosmological history.
 - i) Lecture 15 (Vieregg) — Cosmic Bang in Time-Space; Bodin Ch. 3–4 (Origin and Evolution of the Universe). The expansion and evolution of the universe set the context in which $R_{\star}$ makes sense as a variable.
 - ii) **Sonic mapping:** $R_{\star}$ sets the BPM of the entire synthesizer (range: 40–140 BPM), expressed as a steady quarter-note bass pulse at $\text{Root} \div 4 \text{ Hz}$ (the division by four was somewhat arbitrary, but I did want it to be more of a “motor” note making it lower in pitch than the set frequency allowing for the higher frequency space of the overall sound to be filled by the other variable). It is the only fully periodic/constant sound; all other sounds use randomized placement. This reflects its role as the baseline physical rate from which all other terms are modulated!

2) $f_{\square}$ — Fraction of Stars with Planets (Exoplanet Science)

- a) $f_{\square}$ estimates the fraction of stars that host planetary systems. While we know that planets are abundant today, at the time Drake formulated the equation, it was unknown whether planets were common. Data from the Kepler Space Telescope and microlensing surveys have since shown planets to be nearly ubiquitous. Cassan et al. (2012) estimated that, statistically, every Milky Way star hosts at least one planet. The default is set somewhat arbitrarily to 0.92, reflecting this near-certainty; the range is 0.1–1.0.
 - i) Lecture 6 (Rogers) — A Diversity of Rocky Worlds; Seager (2009); Valencia, Moro-Martinez, and Teske (2024) on exoplanet diversity.
 - ii) **Sonic mapping:** Pluck tones distributed randomly across a 64-step grid. Higher $f_{\square}$ increases note density. The randomized placement within the measure reflects the probabilistic nature of the variable (this will come up in other variables as well).

3) n_e — Habitable Worlds per Star (Planetary Habitability)

- a) n_e is the average number of planets per star system capable of supporting life. Originally defined narrowly as planets in the circumstellar habitable zone (liquid water at the surface), this definition has expanded significantly as discussed in Lecture 7. Conservative estimates for rocky planets in the habitable zone are 0.1–0.4 (Rogers 2015; Kopparapu et al. 2013). Including subsurface ocean worlds (Europa, Enceladus, and Titan) in our own solar system raises the upper bound.

The default is set to 0.4.

- i) Lecture 7 (Graham) — Exoplanet Climate and Habitability; Kasting, Toon, and Pollack (1988) on how climate evolved on terrestrial planets; Lecture 8 (Ciesla) — Planetary Formation.
 - ii) **Sonic mapping:** Sustained pad tones placed randomly across 4 measures. Sustain length is longer than other sounds to represent the geological timescales over which habitability manifests. The knob shifts pitch center, modeling the range of conditions that may or may not support life.
- 4) $f_{\square}$ — Fraction Where Life Arises (Origin of Life / Astrobiology)
- a) $f_{\square}$ is the probability that life emerges on a planet with suitable conditions. Earth's fossil record shows life appeared by approximately 3.5 Ga, within roughly 300 million years of stable surface conditions, a geologically short interval. Whether this implies high probability or is an anthropic sampling artifact is unresolved. The Rare Earth hypothesis argues that $f_{\square}$ is vanishingly small. Szostak's RNA world research (Lecture 16) supports the view that self-replicating chemistry is achievable under plausible prebiotic conditions. The range spans 0.0001 to 1.0; the default is 0.13. We have one confirmed data point.
 - i) Lecture 16 (Szostak) — Life's Origin and the Possibilities for Alien Life; Szostak (2017); Mojarro et al. (2021) on the RNA world on Mars; Lecture 8 (Ciesla) — Bergin (2020) on the chemistry of life.
 - ii) **Sonic mapping:** Short square-wave click sounds placed at randomized positions in the 64-step grid. The density of sound in a measure increases with the knob value. The more stochastic placement reflects the discrete, uncertain nature of abiogenesis. (I asked my DJ friend to take a look at this, and he said he wanted more "stochasticity," which I implemented in this section).
- 5) f_i — Fraction Developing Intelligence (Evolutionary Biology)
- a) f_i estimates the fraction of life-bearing planets where intelligence eventually develops. Intelligence evolved once in approximately 3.8 billion years of Earth's history. The study of convergent evolution (Losos 2017, Lecture 12) suggests that certain traits evolve repeatedly across independent lineages. If it is recursive, symbolic cognition is up for debate. Lecture 11 (Ragsdale) examines cephalopod neural architecture as a case study in how sophisticated cognition can arise from a fundamentally different body plan and nervous system organization. Lecture 13 readings, by Alemseged, discuss the near-extinction of the human lineage approximately 900,000 years ago (Hu et al. 2023), illustrating how contingent human survival was. The default is set to 0.01.
 - i) Course connection: Lectures 2 (Westneat), 3 (London), 4 (Margoliash), 11 (Ragsdale), 12 (Luo), 13 (Alemseged); Losos Ch. 1–3 and Conclusions; Albertin et al. (2015) on the octopus genome; Hu et al. (2023) on the human population bottleneck.
 - ii) **Sonic mapping:** Melody tones with unquantized timing. The notes are placed off the rhythmic grid by a random offset up to one full beat. The knob expands scale complexity from a single root pitch through pentatonic, diatonic, and chromatic modes. The off-grid timing

distinguishes this voice from the physically periodic variables: intelligence does not emerge on a fixed schedule.

6) f^c — Fraction That Communicates (Technosignatures / SETI)

- a) f^c captures the probability that an intelligent civilization produces detectable electromagnetic signals. Humans have broadcast radio waves for approximately 120 years, but modern communication technology is increasingly efficient and low-power, reducing unintentional electromagnetic leakage. Lecture 10 (Tarter) examines the technosignature detection problem, and encourages us to think about what kinds of signals would be detectable, over what distances, and at what frequencies. Bostrom (2008) argues that detecting no signals may itself be informative. The default is 0.1.
 - i) Lecture 10 (Tarter) — Detecting Technosignatures; Margot et al. (2019) on the radio search for technosignatures; Bostrom (2008) on the significance of finding nothing.
 - ii) **Sonic mapping:** High-frequency sine tones at harmonic multiples of the root frequency, placed at unquantized intervals within a two-beat window. Occasionally followed by a secondary ping 0.1 seconds later, modeling signal echoes or repeats. The intermittent, high-frequency, off-grid character reflects the narrowband, fleeting nature of a technosignature. Essentially, this is the sound I would naively associate with a technosignature, which I felt was the best I could do considering the open questions surrounding signature research.

7) L — Civilization Lifetime (Existential Risk)

- a) L is the mean duration for which a civilization remains detectable. It is the most consequential variable: N is directly proportional to L , so a short L (caused by nuclear war, climate collapse, biological risk, or other self-inflicted catastrophe) drives N toward zero regardless of favorable values for the other six terms. Lecture 17 (Holz) addressed existential risk directly; the 2025 Doomsday Clock statement places the clock at 89 seconds to midnight. Drake and Sagan both identified L as the term that turned the equation from a scientific problem into an ethical one. The range spans 100 to 1,000,000 years on a logarithmic scale; the default is 10,000 years.
 - i) Lecture 17 (Holz) — Closer Than Ever to the End of the Earth We Know; 2025 Doomsday Clock Statement; Weininger (2020) Ch. 15; Lecture 1 (Serenio); Frank et al. (2023) on Frank Drake.
 - ii) **Sonic mapping:** A sub-bass sine tone at $\text{Root} \div 8$ Hz (lower frequency than the rest, something constant and looming, can be perceived as eerie or steady), fired once per 64-step cycle (every 4 measures). Sustain length is controlled directly by the knob (2–17 seconds), representing how long the civilization's signal persists before fading.

Finally, the modulation section allows any selected variable's knob value to be swept automatically over time using a low-frequency oscillator (LFO). Several Drake Equation parameters are not static constants but vary over cosmological or geological timescales, so I thought this feature would be a nice touch, especially since many real-life synthesizers and pad loopers have the option to modulate EQ or add (commonly used in audio production) low-pass

filters to their sounds.

$R_{\star}$ has declined from its peak approximately 10 billion years ago as galaxies exhaust their gas supplies. $f_{\square}$ and f_i may be time-dependent if life and intelligence take billions of years to arise and are thus more probable in older stellar systems. L is dynamic if civilizations undergo boom-collapse cycles or if technological risk is a transient phase. The modulation feature allows the listener to hear how a continuously changing parameter shifts both the N estimate and the resulting sound texture, illustrating the non-stationarity of these variables in a way that static knob settings cannot.

The LFO waveform shapes (sine, triangle, sawtooth, reverse sawtooth) represent different modes of change: gradual periodic oscillation, linear ramp and return, asymmetric increase followed by reset, and asymmetric decrease followed by reset.

How does this fit with the overall Golden Record? The Golden Record (1977) was designed to communicate the nature of Earth and humanity to an extraterrestrial intelligence. The Drake Equation Looper inverts this framing: rather than transmitting a message outward, it uses the Drake Equation (the scientific framework for asking whether anyone is out there to receive such a message) as the generative logic for the sound. I think this is apt because it is an auditory and interactive representation of how technology has shaped our more analog tendencies as humans - art, creativity, the propensity to explore and make scientific and technological connections to the arguable subjective world we all perceive around us. I also think audio dials are a perfect format for the Drake Equation, given how this equation, by many of our lecturers, was deemed a bit arbitrary (just a guess). Tuning each knob to a different level allows us to hear the differences that certain variables make to an overall sound that represents n . And finally, the uncertainty encoded in each variable's range is also something I think ought to be communicated to anyone who finds the Golden Record. There is a lack of hubris we've attempted to elucidate in this class that demonstrates that our scientific inquiry is trapped in an anthropocentric view that is also necessary for scientific progress. At the end of the day, the Golden record ought to reflect the human experience, and part of that human experience is admitting that there are things we don't know and that we recognize that we are a tiny part of the universe.

Project Elements Checklist Form

Lecture Number/Topic	Concepts included in the Project	Location in the Project/Essay/Write-Up
Lec 1 (Serenio) ASTRO	Drake Equation structure and purpose; Fermi's paradox as context for N ; Frank Drake's original formulation	All 7 variable pads; About panel overview; N readout
Lec 2 (Westneat) BIO	Taxonomy, tree of life, evolution	Evolution and Tree of Life in Golden Record

Lec 6 (Rogers Lecture and Reading) ASTRO	Diversity of rocky worlds; planet types and habitability; exoplanet survey data (Kepler)	f _☐ pad and About panel entry; default value 0.92 from Kepler data Exoplanetary and planetary data in Golden Record
Lec 7 (Graham Reading) ASTRO	Exoplanet climate; habitable zone definition; Kasting et al. (1988) on terrestrial planet climate	n _e pad; About panel entry on habitable zone and ocean worlds
Lec 10 (Tarter Reading& Lecture) ASTRO	Technosignature detection; narrowband radio search; SETI methodology	f [Ⓢ] pad; sonic mapping as intermittent high-frequency pings; About panel; List of technosignatures on Golden Record
Lec 15 (Vieregg Reading) ASTRO	Cosmic timescales; evolution of the universe; star formation history	R★ pad; modulation section justification (R★ decline over cosmological time) Entire Physics section of the Golden Record
Lec 16 (Szostak Reading) BIO	Origin of life; RNA world hypothesis; prebiotic chemistry; Mars astrobiology	f _☐ pad; About panel entry; Szostak (2017) and Mojarro et al. (2021) cited
Lec 11 (Ragsdale Reading + Lecture) BIO	Cephalopod neural complexity; alternative intelligence; convergent evolution of cognition	f _i pad; About panel entry on non-human intelligence as evidence for f _i range
Lec 12 (Luo Reading) BIO	Convergent evolution; replicated traits across lineages; Losos on evolutionary propensities	f _i pad; About panel discussion of whether intelligence is convergent
Lec 13 BIO (Alemseged Reading)	Human lineage survival; population bottleneck ~900 kya (Hu et al. 2023); contingency of intelligence	f _i pad; About panel History of humans of Golden Record
Lec 17 (Holz) BIG PICTURE	Existential risk; Doomsday Clock; nuclear, climate, and biological risks to civilization lifetime	L pad; About panel; Doomsday Clock statement cited; L as most consequential variable

Collaborator Contributions:

If the project is submitted by a team of more than 1 student, please summarize each collaborators' contributions.

Shared responsibilities: Researching the original Voyager Golden Record—its design, contents, and encoding methodology—to inform the conceptual framing of the New Golden Record project; general website design and structuring.

Ha: I was responsible for most of the Golden Record content. I identified the categories that needed to be covered in order to give potential recipients a broad but sufficiently substantial and diverse impression of life on earth, as well as universal overlaps in math and science and that would give them entry points into understanding our knowledge. I conducted extensive research, using class content as well as resources substantially beyond the scope of the course, in order to compile strong, scientifically/culturally up-to-date, and interactive resources for the Record/website, as well as whittled down broad theories into more simple, easily transmittable and understandable summaries.

Rhea: Designed and built the Drake Equation Looper: all audio engine code (Web Audio API synthesizer voices, 64-step generative scheduler, LFO modulation system), user interface, knob interaction, visual design, and in-app scientific documentation was crafted by Rhea, and once research and tech spec was conducted, Rhea used Claude code to help her build out the actual audio engine. I (Rhea) was able to carry this out because of my music theory and audio production background, knowledge of audio editing techniques, and overall love of electronic music (I have a lot of friends who are DJs). The functionality of the program was entirely designed by me, and being a computer science student, I (Rhea) went through and made sure that there were no hallucinations in the code that had unintended behaviors. Rhea also researched scientific consensus values and ranges for all seven Drake Equation variables, with reference to certain lectures and assigned readings. She also synthesized relevant course lecture material (lectures on exoplanet habitability, origin of life, evolutionary biology, technosignatures, existential risk, and cosmic timescales) to connect each Drake Equation variable to specific scientific topics covered in class. Rhea's research contributions for the Golden Record project (apart from the Looper) covered the following content areas from class readings and lectures: technology (AI, phones, and information theory) [Hao, Sourati, Evans, Lai, Margot], Earth and solar system formation [Chambers, Boyle, Kasting], the historical evolution of life on Earth [Kröger, Mearns, Bostrom, Luo, Hu], and biosignatures [Bergin, Schwieterman & Leung, Seager].

GENERATIVE AI STATEMENT:

All content written in this document and all text you see within the website was written by Ha and I without the help of Generative AI. We conducted research in a document prior to building the website and provided our research to Claude and Gemini to help us turn our tech spec and research into reality. While I don't have all of my chats with chatbots saved (I tend to use guest mode or delete my chats to avoid chatbot personalization effects on my chats - a personal preference given my distaste for sycophancy and conflated understandings of self that come with giving a generative AI chatbot more context). Using AI to code was a choice we made, given the efficiency of chatbots at generating HTML and JavaScript. Rhea is a computer science major, and Ha is well-versed in code, so our ability to edit the code and understand what it does makes us qualified to "vibe code" responsibly. All specifications for the content in each of the Golden Record sections were crafted through our independent research. The tech specifications for the audio engine looper were all defined by us as well. Here is a folder full of all the AI chats I could scrape: sometimes I would post all my notes in and that is much too much content to cut and paste or screenshot so you will only see a fraction of my copy and pasted notes that I ask the bot to not deviate from:

https://drive.google.com/drive/folders/19oA3UmB_RlvTX-aTgYWUO1jgZ2kVJY1?usp=sharing

You can see that some of the things I ask these bots are just styling, and otherwise, it is to translate my notes and uploaded research files/ OG hardcoded HTML naive demo into a more beautiful, easy-to-use site. Let me know if you have any questions. Most chats have been deleted. I try to keep my chat history to less than 5 chats total.

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